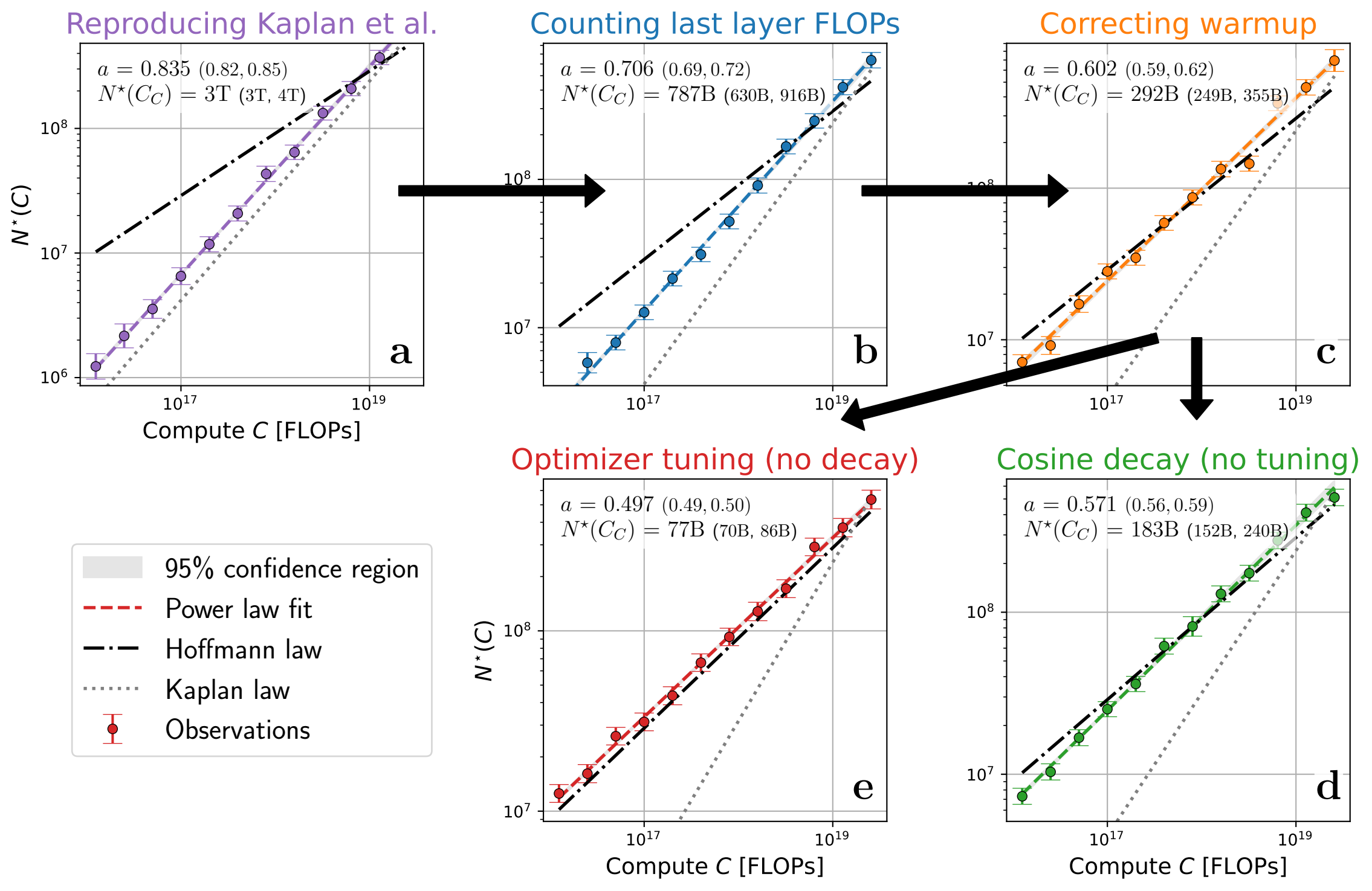

Reproducing Kaplan et al.
$a = 0.835$ (0.82, 0.85)
$N^\star(C_C) = 3\text{T}$ (3T, 4T)
a
Counting last layer FLOPs
$a = 0.706$ (0.69, 0.72)
$N^\star(C_C) = 787\text{B}$ (630B, 916B)
b
Correcting warmup
$a = 0.602$ (0.59, 0.62)
$N^\star(C_C) = 292\text{B}$ (249B, 355B)
c
Optimizer tuning (no decay)
$a = 0.497$ (0.49, 0.50)
$N^\star(C_C) = 77\text{B}$ (70B, 86B)
e
Cosine decay (no tuning)
$a = 0.571$ (0.56, 0.59)
$N^\star(C_C) = 183\text{B}$ (152B, 240B)
d
$N^\star(C)$
Compute C [FLOPs]
95% confidence region
Power law fit
Hoffmann law
Kaplan law
Observations